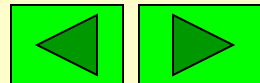


《线性代数与空间解析几何》

第一章 n 阶行列式

1.4 克莱姆法则



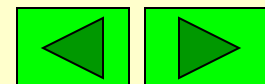
1.4 克莱姆 (Cramer) 法则

下面给出利用 n 阶行列式求解方程个数与未知量个数都是 n 而且系数行列式不为零的线性方程组的求解公式.

定理5 (Cramer 法则) 如果 n 元线性方程组

$$(1) \quad \begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{cases}$$

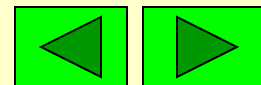
的系数行列式不等于零,



即 $D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \neq 0$

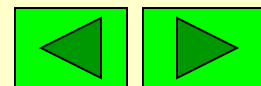
则方程组(1)只有唯一解, 且其解为

$$x_1 = \frac{D_1}{D}, x_2 = \frac{D_2}{D}, \cdots, x_n = \frac{D_n}{D}$$



其中

$$D_j = \begin{vmatrix} a_{11} & \cdots & a_{1j-1} & b_1 & a_{1j+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2j-1} & b_2 & a_{2j+1} & \cdots & a_{2n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nj-1} & b_n & a_{nj+1} & \cdots & a_{nn} \end{vmatrix}$$



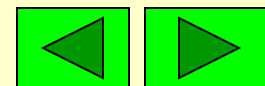
推论1 如果 n 元齐次线性方程组的系数行列式不等于零, 即

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = 0 \end{cases}$$

$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \neq 0$$

则此方程组只有唯一零解, 即

$$x_1 = x_2 = \cdots = x_n = 0.$$

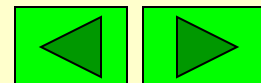


推论2 如果 n 元齐次线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = 0 \end{cases}$$

有非零解, 则系数行列式等于零, 即

$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = 0$$



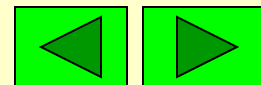
例1 求解线性方程组

$$\begin{cases} x_1 + x_2 + x_3 = 1 \\ x_1 + 2x_2 - x_3 = 0 \\ 3x_1 + 5x_2 + x_3 = 3 \end{cases}$$

解 线性方程组的系数行列式

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ 3 & 5 & 1 \end{vmatrix} = 2 \neq 0$$

所以方程组有唯一解.

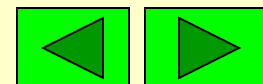


$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ 3 & 5 & 1 \end{vmatrix} = -2 \quad D_2 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 3 & 3 & 1 \end{vmatrix} = 2$$

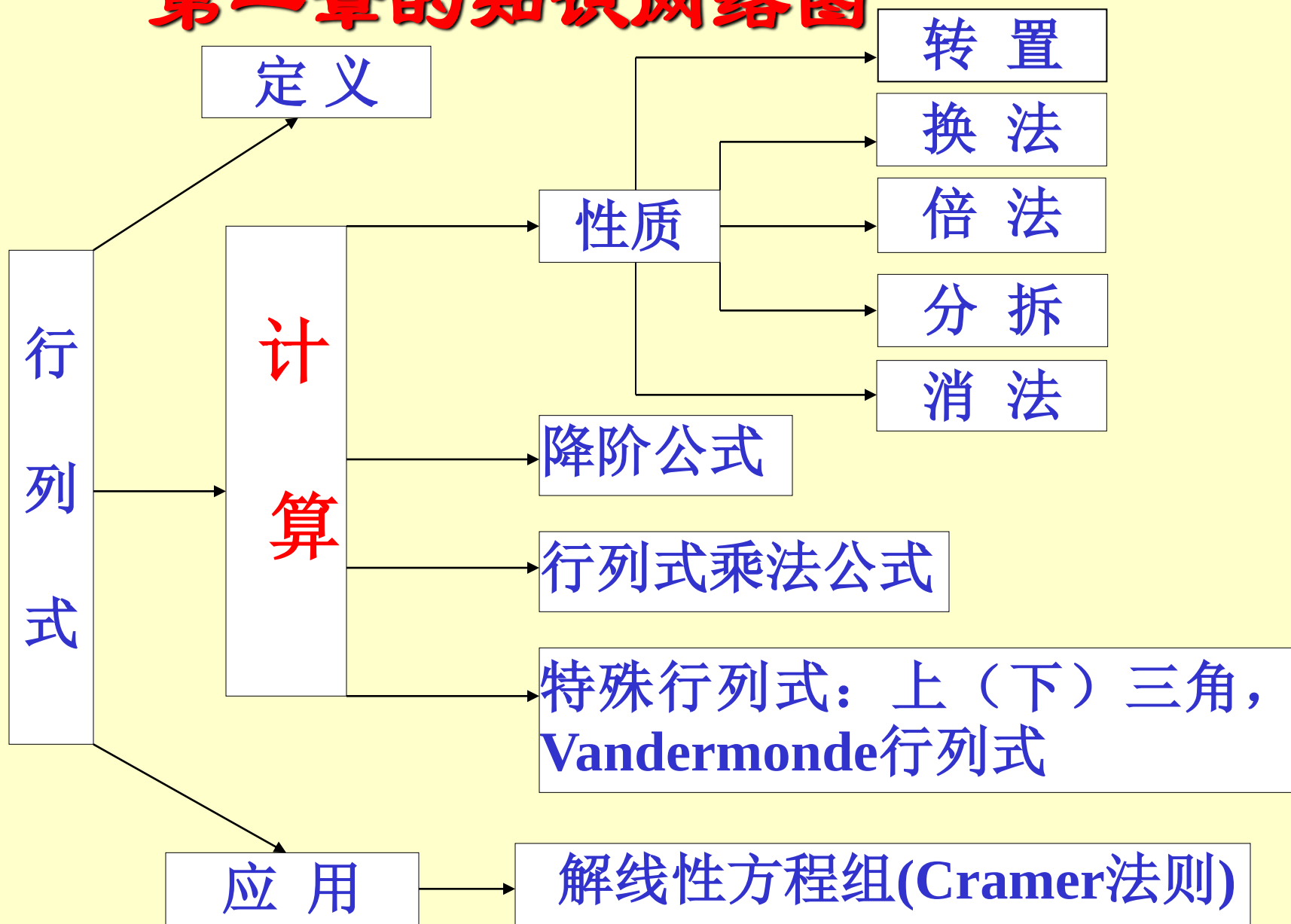
$$D_3 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 3 & 5 & 3 \end{vmatrix} = 2$$

所以方程组的唯一解为

$$x_1 = \frac{D_1}{D} = -1, x_2 = \frac{D_2}{D} = 1, x_3 = \frac{D_3}{D} = 1$$

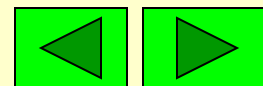


第一章的知识网络图



练习

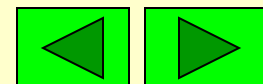
1. 若行列式 D 的某一行元素的代数余子式全是零, 则这个行列式 $D = \underline{0}$.
2. 若4阶行列式 D 的某一行的所有元素及其余子式都相等, 则 $D = \underline{0}$.
3. 在一个 n 阶行列式 D 中, 如果等于零的元素多于 $n^2 - n$ 个, 则 $D = \underline{0}$.



4. 不计算行列式值，利用性质证明

$$\begin{vmatrix} x & x & 2 \\ 2 & x+1 & 3 \\ 3 & 3 & x+1 \end{vmatrix} = (x-1)(x-2)(x+3)$$

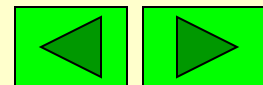
证 令 $f(x) = \begin{vmatrix} x & x & 2 \\ 2 & x+1 & 3 \\ 3 & 3 & x+1 \end{vmatrix}$



由于 $f(x)$ 是 x 的三次多项式, 且

$$f(1) = \begin{vmatrix} 1 & 1 & 2 \\ 2 & 2 & 3 \\ 3 & 3 & 2 \end{vmatrix} = 0, \quad f(2) = \begin{vmatrix} 2 & 2 & 2 \\ 2 & 3 & 3 \\ 3 & 3 & 3 \end{vmatrix} = 0$$

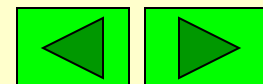
$$f(-3) = \begin{vmatrix} -3 & -3 & 2 \\ 2 & -2 & 3 \\ 3 & 3 & -2 \end{vmatrix} = 0$$



因此有

$$\begin{vmatrix} x & x & 2 \\ 2 & x+1 & 3 \\ 3 & 3 & x+1 \end{vmatrix} = (x-1)(x-2)(x+3)$$

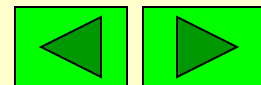
注 x^3 的系数为1.



5

计算行列式

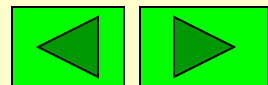
$$D = \begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1-a & 1 & 1 \\ 1 & 1 & 1+b & 1 \\ 1 & 1 & 1 & 1-b \end{vmatrix}$$



解

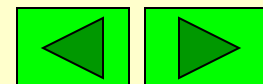
$$D \begin{array}{l} -r_2 + r_1 \\ -r_4 + r_3 \\ \hline \hline \end{array} \begin{vmatrix} a & a & 0 & 0 \\ 1 & 1-a & 1 & 1 \\ 0 & 0 & b & b \\ 1 & 1 & 1 & 1-b \end{vmatrix} = ab \begin{vmatrix} 1 & 1 & 0 & 0 \\ 1 & 1-a & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1-b \end{vmatrix}$$

$$\begin{array}{l} -r_1 + r_2 \\ -r_1 + r_4 \\ \hline \hline \end{array} ab \begin{vmatrix} 1 & 1 & 0 & 0 \\ 0 & -a & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1-b \end{vmatrix} \begin{array}{l} -r_3 + r_4 \\ \hline \hline \end{array} ab \begin{vmatrix} 1 & 1 & 0 & 0 \\ 0 & -a & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -b \end{vmatrix} = a^2 b^2$$



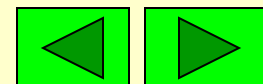
6. 计算行列式

$$D = \begin{vmatrix} x_1 - m & x_2 & x_3 & \cdots & x_n \\ x_1 & x_2 - m & x_3 & \cdots & x_n \\ x_1 & x_2 & x_3 - m & \cdots & x_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_1 & x_2 & x_3 & \cdots & x_n - m \end{vmatrix}$$



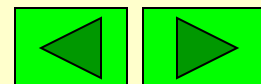
解

$$\begin{aligned}
 D &= \begin{vmatrix} -m + \sum_{i=1}^n x_i & x_2 & \cdots & x_n \\ -m + \sum_{i=1}^n x_i & x_2 - m & \cdots & x_n \\ \vdots & \vdots & & \vdots \\ -m + \sum_{i=1}^n x_i & x_2 & \cdots & x_n - m \end{vmatrix} \\
 &= \left(-m + \sum_{i=1}^n x_i\right) \begin{vmatrix} 1 & x_2 & \cdots & x_n \\ 1 & x_2 - m & \cdots & x_n \\ \vdots & \vdots & & \vdots \\ 1 & x_2 & \cdots & x_n - m \end{vmatrix}
 \end{aligned}$$



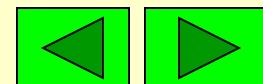
$$= (-m + \sum_{i=1}^n x_i) \begin{vmatrix} 1 & x_2 & \cdots & x_n \\ 0 & -m & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & -m \end{vmatrix}$$

$$= (-m + \sum_{i=1}^n x_i) (-m)^{n-1}$$



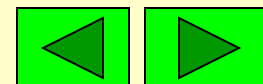
7. 计算行列式: $(x_i \neq a_i)$

$$D = \begin{vmatrix} x_1 & a_2 & a_3 & a_4 \\ a_1 & x_2 & a_3 & a_4 \\ a_1 & a_2 & x_3 & a_4 \\ a_1 & a_2 & a_3 & x_4 \end{vmatrix}$$

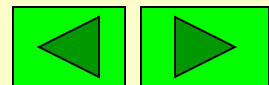


解 用加边法

$$D = \left| \begin{array}{ccccc} 1 & a_1 & a_2 & a_3 & a_4 \\ 0 & x_1 & a_2 & a_3 & a_4 \\ 0 & a_1 & x_2 & a_3 & a_4 \\ 0 & a_1 & a_2 & x_3 & a_4 \\ 0 & a_1 & a_2 & a_3 & x_4 \end{array} \right|$$



$$\underline{\underline{-r_1 + r_i (i = 2, 3, 4, 5)}} \left| \begin{array}{ccccc} 1 & a_1 & a_2 & a_3 & a_4 \\ -1 & x_1 - a_1 & 0 & 0 & 0 \\ -1 & 0 & x_2 - a_2 & 0 & 0 \\ -1 & 0 & 0 & x_3 - a_3 & 0 \\ -1 & 0 & 0 & 0 & x_4 - a_4 \end{array} \right|$$

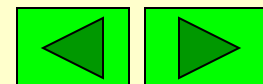


$$\begin{aligned}
&= \begin{vmatrix} 1 + \sum_{i=1}^4 \frac{a_i}{x_i - a_i} & a_1 & a_2 & a_3 & a_4 \\ 0 & x_1 - a_1 & 0 & 0 & 0 \\ 0 & 0 & x_2 - a_2 & 0 & 0 \\ 0 & 0 & 0 & x_3 - a_3 & 0 \\ 0 & 0 & 0 & 0 & x_4 - a_4 \end{vmatrix} \\
&= \left(1 + \sum_{i=1}^4 \frac{a_i}{x_i - a_i}\right) \prod_{i=1}^4 (x_i - a_i)
\end{aligned}$$

8. 已知 $D = \begin{vmatrix} 1 & 3 & 0 & 1 \\ 3 & 0 & 1 & 4 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 0 \end{vmatrix}$

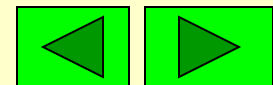
计算 (1) $A_{31} + A_{32} + A_{33} + A_{34}$

(2) $M_{31} + M_{32} + M_{33} + M_{34}$



解 $A_{31} + A_{32} + A_{33} + A_{34} = \begin{vmatrix} 1 & 3 & 0 & 1 \\ 3 & 0 & 1 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{vmatrix} = -3.$

$$M_{31} + M_{32} + M_{33} + M_{34} \\ = A_{31} - A_{32} + A_{33} - A_{34} \\ = \begin{vmatrix} 1 & 3 & 0 & 1 \\ 3 & 0 & 1 & 4 \\ 1 & -1 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{vmatrix} = -25.$$

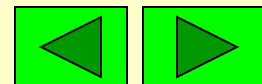


9. 设有四阶行列式 $D = \begin{vmatrix} 3 & 7 & 9 & 7 \\ 4 & 8 & 5 & 6 \\ 3 & 0 & 4 & 9 \\ 1 & 4 & 7 & 2 \end{vmatrix}$

设 $a = 4A_{41} + 8A_{42} + 5A_{43} + 6A_{44}$, 则 a 的值为:

(A) -2; (B) -1; (C) 0; (D) 2.

分析: a 相当于第2行的元素乘上的第4行的代数余子式, 根据行列式的性质, 应该为0, 答案为(C).

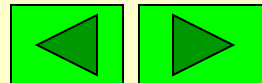


10. 设 $f(x)$ 是一个次数不大于 $n-1$ 的一元多项式, 如果存在 n 个互不相同的数 $x_1, x_2, \dots, x_n$ 使 $f(x_i) = 0, i = 1, 2, \dots, n$,
证明: $f(x) = 0$

证 设 $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1}$

其中 a_i 待定,

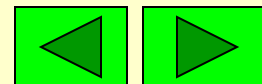
由已知 $f(x_i) = 0, i = 1, 2, \dots, n$



得

$$\begin{cases} a_0 + a_1 x_1 + a_2 x_1^2 + \cdots + a_{n-1} x_1^{n-1} = 0 \\ a_0 + a_1 x_2 + a_2 x_2^2 + \cdots + a_{n-1} x_2^{n-1} = 0 \\ \dots\dots\dots \\ a_0 + a_1 x_n + a_2 x_n^2 + \cdots + a_{n-1} x_n^{n-1} = 0 \end{cases}$$

这是关于 $a_0, a_1, a_2, \dots, a_{n-1}$ 的 n 元一次线性方程组, 其系数行列式



$$D = \begin{vmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{vmatrix}$$

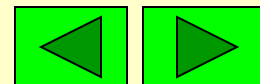
$$= \prod_{1 \leq j < i \leq n} (x_i - x_j) \neq 0$$

所以方程组只有唯一零解,即

$$a_0 = a_1 = a_2 = \cdots = a_{n-1} = 0$$

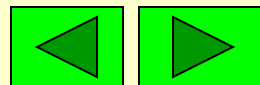
故

$$f(x) = 0$$



《线性代数与空间解析几何》

第二章 矩阵



本章的主要内容

- 矩阵的概念及运算
- 可逆矩阵*
- 矩阵的初等变换与初等阵*
- 矩阵的秩
- 分块矩阵

2.1 矩阵的概念

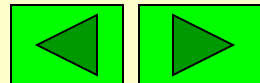
2.1.1. 矩阵的概念

1 矩阵的定义

由 $m \times n$ 个数排成的 m 行、 n 列的矩形数表

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

称为 $m \times n$ 矩阵.



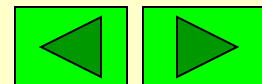
简记为 $A_{m \times n} = (a_{ij})_{m \times n}$

当 $m=n$ 时称为 n 阶方阵.

- 矩阵同形 $\Leftrightarrow$ 它们行数和列数相同.
- 矩阵相等 $\Leftrightarrow$ 它们同形且对应元素相等.
- 方阵的行列式: $|A| = |a_{ij}|_{n \times n}$ 或 $\det A$.

2. 特殊矩阵

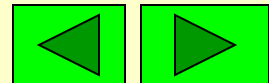
- 零矩阵: $0_{m \times n}, 0$



- 对角矩阵:
$$\begin{pmatrix} a_{11} & & \\ & a_{22} & \\ & & \ddots \\ & & & a_{nn} \end{pmatrix}$$

- 单位矩阵: E, I 或 $E_n = \begin{pmatrix} 1 & & \\ & 1 & \\ & & \ddots \\ & & & 1 \end{pmatrix}$

- 数(标)量矩阵: $A = \begin{pmatrix} k & & \\ & k & \\ & & \ddots \\ & & & k \end{pmatrix}$



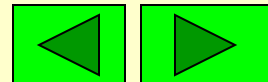
- 上三角矩阵:
$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{pmatrix}$$

- 下三角矩阵:
$$\begin{pmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

- 行矩阵: $(a_1 \ a_2 \ \cdots \ a_n)$
- 列矩阵:
$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

2.2 矩阵运算

- 矩阵的线性运算
- 矩阵的乘法运算
- 方阵的幂及
行列式的乘法公式
- 矩阵的转置



2.2.1 矩阵的加法: (A 与 B 要同形).

● 加法: $A = (a_{ij})_{m \times n}$ $B = (b_{ij})_{m \times n}$

$$A + B = C = (c_{ij})_{m \times n} = (a_{ij} + b_{ij})_{m \times n}$$

负矩阵: $-A = -(a_{ij})_{m \times n} = (-a_{ij})_{m \times n}$

减法: $A - B = A + (-B)$

● 运算性质:

$$A + B = B + A, (A + B) + C = A + (B + C)$$

$$A + \mathbf{0} = A, A + (-A) = \mathbf{0}$$

2.2.2 数乘

$$kA = k(a_{ij})_{m \times n} = (ka_{ij})_{m \times n}$$

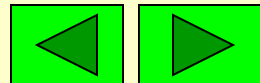
● 运算性质:

$$k(lA) = (kl)A$$

$$k(A + B) = kA + kB$$

$$(k + l)A = kA + lA$$

$$1A = A, \quad 0A = 0$$



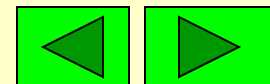
2.2.3 矩阵乘法

● $A = (a_{ij})_{m \times s}$, $B = (b_{ij})_{s \times n}$, 则 $C = AB = (c_{ij})_{m \times n}$

其中 $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{is}b_{sj}$

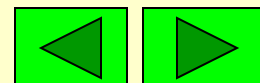
$$= \sum_{k=1}^s a_{ik}b_{kj}, i = 1, \cdots, m; j = 1, \cdots, n.$$

$$\begin{pmatrix} \cdots & \cdots & \cdots & \cdots \\ a_{i1} & a_{i2} & \cdots & a_{is} \\ \cdots & \cdots & \cdots & \cdots \end{pmatrix} \begin{pmatrix} \vdots & b_{1j} & \vdots \\ \vdots & b_{2j} & \vdots \\ \vdots & \vdots & \vdots \\ \vdots & b_{sj} & \vdots \end{pmatrix} = \begin{pmatrix} \cdots & \vdots & \cdots \\ \cdots & c_{ij} & \cdots \\ \cdots & \vdots & \cdots \end{pmatrix}$$



总结如下:

- **可乘原则:** 前列数=后行数.
- **乘积元素:** c_{ij} 是 A 的第 i 行的元素与 B 的第 j 列对应元素乘积之和.
- **乘积阶数:** AB 阶数为前行数 $\times$ 后列数.



● 运算性质: (A 是 $m \times n$ 的矩阵)

$$(1) \mathbf{0}_{p \times m} A = \mathbf{0}_{p \times n}, A \mathbf{0}_{n \times q} = \mathbf{0}_{m \times q}$$

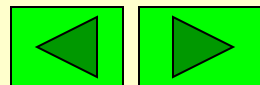
$$(2) E_m A = A, A E_n = A$$

$$(3) A(BC) = (AB)C$$

$$(4) A(B + C) = AB + AC$$

$$(B + C)A = BA + CA$$

- 学习矩阵运算, 尤其要注意其不具备什么熟知的运算规律. 特别是乘法运算.

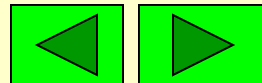


例1 设 $A = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & -1 \\ 0 & 3 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$

求 AB .

解 $\begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & -1 \\ 0 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \times 1 + 3 \times 0 + 1 \times 2 \\ 1 \times 1 + 2 \times 0 + (-1) \times 2 \\ 0 \times 1 + 3 \times 0 + 1 \times 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$

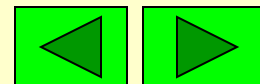
注意: 在这个例子中 BA 无意义.



例2 $A = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad B = (b_1 \quad b_2)$

则 $AB = \begin{pmatrix} a_1 b_1 & a_1 b_2 \\ a_2 b_1 & a_2 b_2 \end{pmatrix}, BA = (b_1 a_1 + b_2 a_2)$

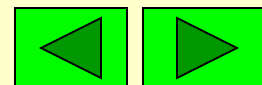
注意：在这个例子中，虽然 AB 与 BA 均有意义，但是 AB 是 2×2 矩阵，而 BA 是 1×1 矩阵。



例3 设 $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

则 $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $BA = \begin{bmatrix} 2 & 2 \\ -2 & -2 \end{bmatrix}$

注意: (1) AB 与 BA 是同阶方阵, 但 AB 不等于 BA . (2) 虽然 A, B 都是非零矩阵, 但是 $AB = 0$.



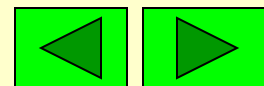
例4 设 $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 \\ -2 & 1 \end{bmatrix}$, $C = \begin{bmatrix} -7 & 1 \\ 1 & 2 \end{bmatrix}$,

求 AB 及 AC .

解 $AB = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 5 \\ -10 & 10 \end{bmatrix},$

$$AC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -7 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -5 & 5 \\ -10 & 10 \end{bmatrix}.$$

注意: 虽然 A 不是零矩阵, 而且 $AB=AC$, 但是 B 不等于 C . 这说明消去律不成立!



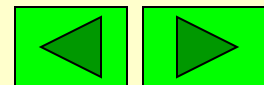
总结一下矩阵乘法的一些反常性质:

- 未必满足交换律: $AB \neq BA$
- 未必满足消去律: $AB = AC \not\Rightarrow B = C$
- 可能有零因子: $AB = 0 \not\Rightarrow A = 0$ 或 $B = 0$

$$A \neq 0 \text{ 且 } B \neq 0 \not\Rightarrow AB \neq 0$$

 如果 $AB=BA$, 则称 A 与 B 可交换.

 学习矩阵理论, 尤其要注意反常性质!



预 习2.2----- 2.3

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预 习 2.2-2.3

<http://www.dreamstime.com/stock-image>



矩阵理论引言

● 历史

- ★ 矩阵思想源于求解线性方程组
- ★ 矩阵思想产生于中国（朱世杰等）
- ★ Matrix概念产生于英国 (Sylvester)

● 应用

- ★ 数学(计算数学)各分支
- ★ 信息处理•系统控制•工程技术等

